Advanced General Relativity

A course on the fundamentals of

Gravitation and Spacetime

Black Holes – Classical and Quantum

Inflation and the Origin of cosmic Structure

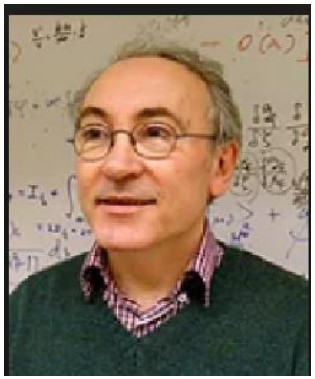
Gravitational Waves – A new window for exploration

Roberto Emparan



General Formalism
Black Holes
Quantum Gravity

Jaume Garriga



Relativistic Cosmology
Inflation
Cosmological perturbations

Ruxandra Bondarescu



Gravitational wave astronomy
GW from black holes and neutron stars
From the stellar graveyard to the universe at large

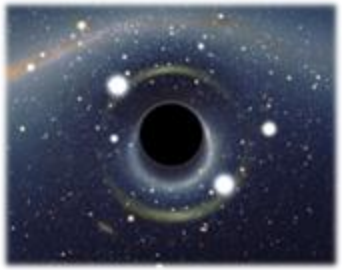
Pre-requisite: Introductory course to General Relativity

What will you learn?

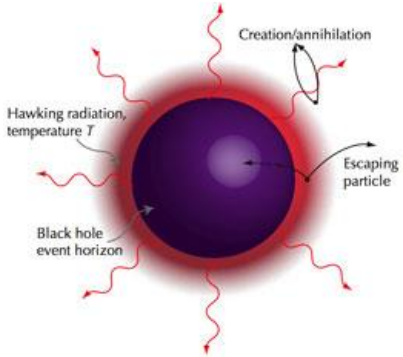




- Advanced techniques in GenRel
  - Lagrangian formulation, Causal Structure (Horizons)...



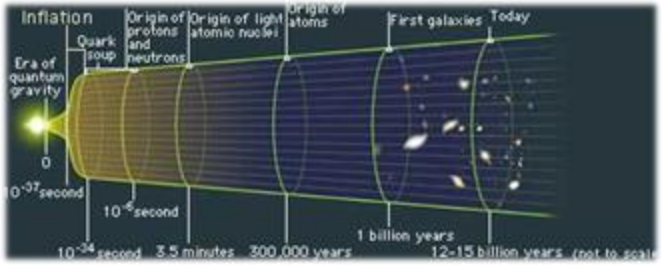
- Black Holes
  - Classical theory
  - Quantum effects – Hawking radiation
  - Black hole thermodynamics



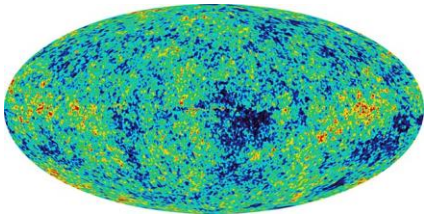
- Basic notions about
  Quantum theory of Gravity



What will you learn?

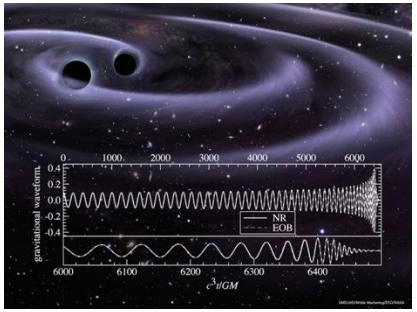


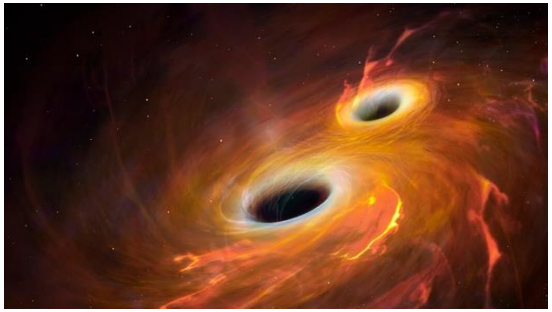
- Relativistic Cosmology. Global structure.
- Cosmological perturbation theory
- Inflation as a theory of initial conditions
  - Quantum origin of primordial perturbations
  - CMB & LSS

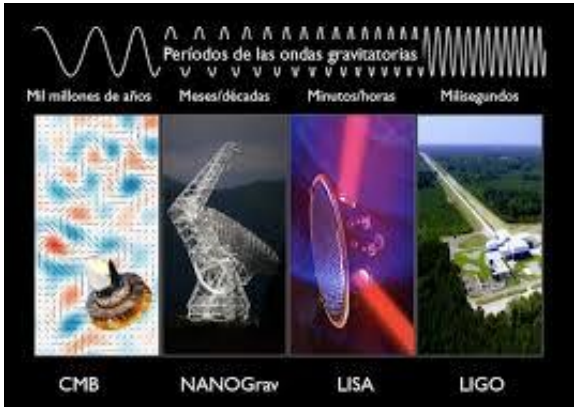


What will you learn?

- Opening the window to gravitational wave astronomy
- The binary inspiral.
- Colliding black holes and colliding neutron stars
- Relativistic asteroseismology
- Mining data and catching gravitational waves
- Learning about the universe at large







What is the course like?

- Theoretical – be ready for math
- On blackboard – we'll go into details
- Homework (weekly)

  complement to lectures

  essential practice

  evaluation*

* alternatively, exam if requested